Printix MCP — Permission Matrix

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Comprehensive list of MCP tools and which roles can invoke them

- Total tools tagged: **127**
- Generated for Printix MCP v7.2.26
- Source of truth: `src/permissions.py` (TOOL_SCOPES, ROLE_SCOPES)

Role-to-scope summary

Role	mcp:self	mcp:read	mcp:audit	mcp:write	mcp:system
End User	✓	—	—	—	—
Helpdesk	✓	✓	—	—	—
Admin	✓	✓	✓	✓	✓
Auditor	—	✓	✓	—	—
Service Acct.	—	—	—	—	—

Scope `mcp:self` — Own data only

Tools that act on the caller's own data — print own jobs, look up own status, ask self-introspection questions.

Allowed roles: End User, Helpdesk, Admin

Tool count: 8

#	Tool
1	`printix_explain_error`
2	`printix_generate_id_code`
3	`printix_my_role`
4	`printix_print_self`
5	`printix_session_print`
6	`printix_status`
7	`printix_suggest_next_action`
8	`printix_whoami`

Scope `mcp:read` — Read-only across the tenant

Tools that list, get, query, search, describe, resolve, or report — no mutations.

Allowed roles: Helpdesk, Admin, Auditor

Tool count: 77

#	Tool
1	printix_capture_status
2	printix_card_audit
3	printix_compare_periods
4	printix_cost_by_department
5	printix_decode_card_value
6	printix_describe_capture_profile
7	printix_describe_user_print_pattern
8	printix_diagnose_user
9	printix_find_orphaned_mappings
10	printix_find_user
11	printix_get_card_details
12	printix_get_card_profile
13	printix_get_feature_request
14	printix_get_group
15	printix_get_group_members
16	printix_get_job
17	printix_get_network
18	printix_get_network_context
19	printix_get_printer
20	printix_get_queue_context
21	printix_get_report_template
22	printix_get_site
23	printix_get_snmp_config
24	printix_get_snmp_context
25	printix_get_user
26	printix_get_user_card_context
27	printix_get_user_groups
28	printix_get_workstation
29	printix_inactive_users
30	printix_jobs_stuck
31	printix_list_admins
32	printix_list_backups
33	printix_list_capture_profiles
34	printix_list_card_profiles
35	printix_list_cards
36	printix_list_cards_by_tenant
37	printix_list_design_options
38	printix_list_feature_requests
39	printix_list_groups
40	printix_list_jobs
41	printix_list_networks
42	printix_list_printers
43	printix_list_report_templates

#	Tool
44	`printix_list_schedules`
45	`printix_list_sites`
46	`printix_list_snmp_configs`
47	`printix_list_timebombs`
48	`printix_list_users`
49	`printix_list_workstations`
50	`printix_natural_query`
51	`printix_network_printers`
52	`printix_permission_matrix`
53	`printix_preview_report`
54	`printix_print_history_natural`
55	`printix_print_trends`
56	`printix_printer_health_report`
57	`printix_query_anomalies`
58	`printix_query_any`
59	`printix_query_cost_report`
60	`printix_query_print_stats`
61	`printix_query_top_printers`
62	`printix_query_top_users`
63	`printix_query_trend`
64	`printix_quota_guard`
65	`printix_reporting_status`
66	`printix_resolve_printer`
67	`printix_resolve_recipients`
68	`printix_search_card`
69	`printix_search_card_mappings`
70	`printix_site_summary`
71	`printix_sso_status`
72	`printix_suggest_profile`
73	`printix_tenant_summary`
74	`printix_top_printers`
75	`printix_top_users`
76	`printix_transform_card_value`
77	`printix_user_360`

Scope `mcp:audit` — Audit log

Read access to the structured audit trail (Art. 30).

Allowed roles: Admin, Auditor

Tool count: 1

#	Tool
1	`printix_query_audit_log`

Scope `mcp:write` — Mutations

Create / update / delete operations on Printix entities and configurations.

Allowed roles: Admin

Tool count: 35

#	Tool
1	`printix_bulk_import_cards`
2	`printix_card_enrol_assist`
3	`printix_change_job_owner`
4	`printix_complete_upload`
5	`printix_create_group`
6	`printix_create_network`
7	`printix_create_site`
8	`printix_create_snmp_config`
9	`printix_create_user`
10	`printix_delete_card`
11	`printix_delete_group`
12	`printix_delete_job`
13	`printix_delete_network`
14	`printix_delete_report_template`
15	`printix_delete_schedule`
16	`printix_delete_site`
17	`printix_delete_snmp_config`
18	`printix_delete_user`
19	`printix_offboard_user`
20	`printix_onboard_user`
21	`printix_print_to_recipients`
22	`printix_quick_print`
23	`printix_register_card`
24	`printix_run_report_now`
25	`printix_save_report_template`
26	`printix_schedule_report`
27	`printix_send_test_email`
28	`printix_send_to_capture`
29	`printix_send_to_user`
30	`printix_submit_job`
31	`printix_sync_entra_group_to_printix`
32	`printix_update_network`
33	`printix_update_schedule`

#	Tool
34	`printix_update_site`
35	`printix_welcome_user`

Scope `mcp:system` — System administration

Backups, demo data, time-bomb engine, sensitive system commands.

Allowed roles: Admin

Tool count: 6

#	Tool
1	`printix_create_backup`
2	`printix_defuse_timebomb`
3	`printix_demo_generate`
4	`printix_demo_rollback`
5	`printix_demo_setup_schema`
6	`printix_demo_status`

Notes

- Tools not listed in this document fall back to `mcp:write` by default. This is intentional safe-by-default behaviour: any tool added to the server without an explicit scope tag remains admin-only until an operator categorises it.
- The Service Account role has no implicit scopes; permissions for non-human tokens are whitelisted explicitly per token.
- The MCP introspection tool `printix_my_role` is always available regardless of role — every user can ask their AI assistant *"what can I do?"* and get a structured answer back.
- Denied calls are recorded with `action='mcp_permission_denied'` in the `audit_log` table for compliance review.